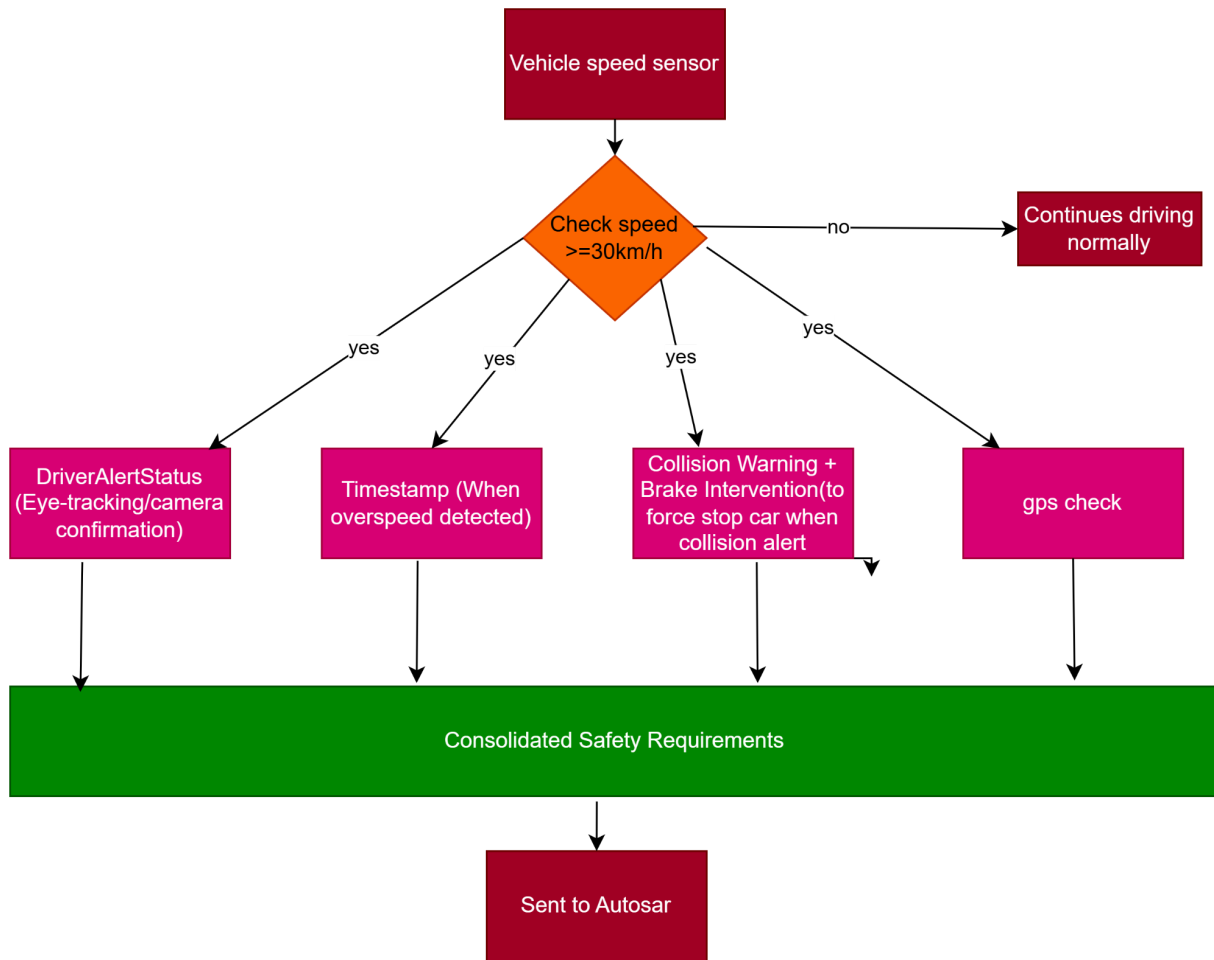




Process Logic & Decision Points

Activation Threshold

- **Trigger Mechanism:** Vehicle Speed Sensor (VSS).
- **Threshold Value:** 30km/h
- **Behavior:** * If **No**: The system allows the vehicle to continue driving normally without advanced data logging.
 - If **Yes**: The system concurrently activates four primary data/intervention streams.

Concurrent Data Streams

Upon activation, the following subsystems are polled:

1. **Driver Alert Status:** Utilizes eye-tracking and camera-based confirmation to ensure driver attentiveness.
2. **Temporal Logging:** Records a high-precision timestamp the moment the speed threshold or overspeed condition is met.

3. **Active Safety (Collision Warning/Brake Intervention):** Monitors for immediate obstacles, includes the capability to trigger autonomous emergency braking (AEB) to force a stop if a collision is imminent.
4. **Spatial Logging (GPS):** Captures real-time geographical coordinates to provide context for the safety event.

AUTOSAR Integration

The final consolidated data packet is formatted and transmitted to the **AUTOSAR (Automotive Open System Architecture)** stack. This allows for standardized communication between different Electronic Control Units (ECUs) and ensures the data can be utilized by higher-level vehicle diagnostic or telematics systems.